

SAP Capstone Project

Order-to-Cash (O2C)

Complete Sales Cycle

SAP SD Module | End-to-End Scenario

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1. Introduction

SAP (Systems, Applications, and Products in Data Processing) is a globally leading enterprise resource planning (ERP) software that integrates every facet of a business. The SAP SD (Sales and Distribution) module is one of the core functional modules of SAP ERP, responsible for handling all business processes associated with sales, shipping, and billing. It tightly integrates with other SAP modules such as MM (Materials Management), FI (Financial Accounting), and PP (Production Planning) to deliver a seamless end-to-end enterprise workflow.

This capstone project focuses on the Order-to-Cash (O2C) business process — one of the most critical and comprehensive end-to-end scenarios in SAP SD. The O2C cycle covers every step from the moment a customer places an inquiry to the point where payment is received and posted. Understanding and executing this cycle in SAP is a fundamental competency for any SAP SD consultant.

2. Problem Statement

In many businesses, the sales process involves multiple departments — sales, warehouse, logistics, and finance — each working in silos, leading to delays, errors in order fulfillment, and revenue leakage. The challenge is to integrate all these functions into a unified, automated, and transparent workflow.

This project aims to demonstrate how SAP SD's Order-to-Cash cycle can be configured and executed for a fictitious company, **TechNova Pvt. Ltd.**, to streamline its complete sales process — from pre-sales activities like inquiry and quotation to post-sales activities like invoice generation and payment collection — thereby eliminating redundancies and improving operational efficiency.

Fictitious Company Profile:

Company Name	TechNova Pvt. Ltd.
Industry	Electronics & Consumer Goods
Company Code	TN01
Sales Org.	SO01 – TechNova Sales Organization
Distribution Ch.	DC01 – Direct Sales
Division	DV01 – Electronics
Plant	PL01 – TechNova Warehouse, Bengaluru

3. Solution & Features of the O2C Cycle

The Order-to-Cash process in SAP SD is a structured, multi-step business cycle. Each step is executed through specific SAP transactions and generates subsequent documents. Below is the complete step-by-step execution of the O2C process for TechNova Pvt. Ltd.:

Step 1	Pre-Sales – Customer Inquiry (Transaction: VA11) The cycle begins when a customer contacts TechNova to inquire about product availability, pricing, and delivery timelines. In SAP, this is captured using transaction VA11 to create an Inquiry document. The inquiry records the customer details, product interest, and tentative quantity. This step is optional but provides a formal record of customer interest.
Step 2	Quotation Creation (Transaction: VA21) Based on the inquiry, the sales team prepares a formal quotation using VA21. The quotation includes pricing (determined by Pricing Conditions – PR00, K007, MWST), validity dates, and terms of delivery. The quotation is shared with the customer. SAP automatically references the inquiry in the quotation, maintaining document flow traceability.
Step 3	Sales Order Creation (Transaction: VA01) Upon customer acceptance of the quotation, a Sales Order is created in VA01 referencing the quotation. The sales order is the key document in O2C — it captures ordered material, quantity, pricing, delivery schedule, and payment terms. SAP performs availability check (ATP) and credit limit check automatically at this stage.
Step 4	Delivery Document Creation (Transaction: VL01N) Once the sales order is confirmed, an outbound delivery document is created via VL01N. This initiates the logistics process. The system picks the goods from the warehouse (Pick – LT0A), packs them, and records the actual quantity to be shipped. Goods Issue (GI) is posted, which reduces inventory and triggers an accounting entry: Debit – COGS, Credit – Inventory.
Step 5	Billing / Invoice Creation (Transaction: VF01) After goods issue, an invoice (billing document) is created via VF01. The billing document is based on the delivery and reflects the final amounts. It triggers the FI integration: Debit – Customer A/R, Credit – Revenue Account. The invoice is then printed/emailed to the customer.

Step 6**Payment Receipt (Transaction: F-28 / FB05)**

The final step is payment collection. When the customer pays, the payment is posted in SAP FI using transaction F-28 (incoming payment). The open item (invoice) is cleared against the payment, completing the O2C cycle. The accounts receivable balance is reduced to zero for that transaction.

4. Technical Stack & SAP Configuration

4.1 SAP Technology Components Used

Component	Details
SAP Module	SAP SD (Sales & Distribution)
SAP Version	SAP ECC 6.0 / SAP S/4HANA
Integration	SAP FI (Finance), SAP MM (Materials Mgmt), SAP WM (Warehouse Mgmt)
Org. Structure	Company Code, Sales Org, Distribution Channel, Division, Plant
Master Data	Customer Master, Material Master, Pricing Conditions, Info Records
Key Transactions	VA11, VA21, VA01, VL01N, VF01, F-28
Configuration Tool	SAP IMG (SPRO) – Implementation Guide
Reporting	SAP Standard Reports: VA05, VF05, VL06O, ME2M

4.2 Enterprise Structure Configuration (SPRO Path)

- Define Company Code: SPRO > Enterprise Structure > Definition > Financial Accounting > Define Company Code
- Define Sales Organization: SPRO > Enterprise Structure > Definition > Sales and Distribution > Define Sales Organization
- Define Distribution Channel & Division: Maintained under SD Enterprise Structure
- Assign Sales Org to Company Code: SPRO > Enterprise Structure > Assignment > SD
- Define Plant and Storage Location: SPRO > Enterprise Structure > Definition > Logistics General
- Maintain Sales Area: Combination of Sales Org + Distribution Channel + Division

4.3 Master Data Created

Master Data	Transaction	Key Fields
Customer Master	XD01	Account Group, Reconciliation Account, Sales Area data, Payment
Material Master	MM01	Material Type, Sales Org views, MRP, Pricing, Plant data
Pricing Conditions	VK11	PR00 (Price), K007 (Customer Discount), MWST (Tax)
Credit Limit	FD32	Credit Control Area, Credit Limit Amount

5. Document Flow & Process Screenshots

SAP automatically maintains a Document Flow for every transaction in the O2C cycle. This allows users to trace any document back to its origin and forward to its completion. Below is the complete document flow for TechNova's O2C scenario:

Document Type	SAP Document No.	Transaction	Status
Customer Inquiry	10000001	VA11/VA12	Completed
Sales Quotation	20000001	VA21/VA22	Completed
Sales Order	30000001	VA01/VA02	Completed
Outbound Delivery	80000001	VL01N/VL02N	Goods Issued
Transfer Order	TR00001	LT0A	Confirmed
Billing Document	90000001	VF01/VF02	Posted to FI
Accounting Doc.	1400000001	FB03	Cleared

5.1 Key Process Screens (Description)

Sales Order (VA01)	The sales order screen captures customer number (e.g., CUST001), material (e.g., MAT-LAPTOP-01), order quantity (50 units), requested delivery date, and pricing. The order value auto-calculates via pricing procedure RVAA01.
Delivery Document (VL01N)	The delivery document shows the picking quantity, storage location, and goods issue date. After picking confirmation, Goods Issue is posted, reducing stock by 50 units and posting a COGS entry.
Billing Document (VF01)	The billing document mirrors the delivery data and generates the customer invoice. Net value, tax (GST), and total amount payable are clearly shown. The FI document is automatically created upon saving.
Accounting Entry (FB03)	FI document shows: Dr. Customer A/R – ₹5,90,000 / Cr. Revenue – ₹5,00,000 / Cr. GST Payable – ₹90,000. Upon payment receipt (F-28), the A/R is cleared.

6. Unique Points & Value Additions

This project goes beyond standard textbook-level SAP configuration by incorporating real-world scenarios and best practices observed in enterprise SAP implementations:

Credit Management Integration:

The sales order creation checks the customer credit limit in real time. If TechNova's customer exceeds their credit limit of ₹10,00,000, the order is automatically blocked (Credit Block) and requires release via FD32/VKM3. This prevents revenue loss due to bad debt.

Pricing Procedure (RVAA01):

A custom pricing procedure was configured with multiple condition types: Base Price (PR00), Customer-specific Discount (K007), Freight Charges (KF00), and GST Tax (MWST). The net value is auto-calculated through the pricing procedure, eliminating manual pricing errors.

Output Determination:

Output types (RD00 – Invoice, BA00 – Order Confirmation) were configured to auto-generate PDF outputs. These can be emailed directly from SAP to the customer, demonstrating real-world document management within SAP.

Availability Check (ATP Logic):

The sales order performs an Available-to-Promise (ATP) check against plant inventory. If stock is insufficient, SAP proposes a confirmed delivery date based on planned goods receipt, ensuring realistic delivery commitments to customers.

Complete FI-SD Integration:

Every billing document automatically generates an FI accounting entry without any manual intervention, demonstrating the power of SAP's integrated architecture. Revenue recognition is automated and compliant with financial regulations.

7. Future Improvements

While the current project demonstrates the core O2C cycle comprehensively, the following enhancements could further improve the solution in a real-world context:

- **SAP S/4HANA Migration:** Migrate the current ECC-based O2C process to SAP S/4HANA, leveraging HANA's in-memory computing for real-time analytics and simplified data models (e.g., ACDOCA universal journal table replacing multiple FI tables).
- **Customer Portal Integration (SAP Commerce Cloud):** Allow customers to self-serve by placing orders directly via a web portal, which automatically creates sales orders in SAP backend through standard APIs, reducing manual order entry effort.
- **SAP BW/4HANA Reporting Dashboard:** Build an executive sales dashboard showing KPIs like Order-to-Cash cycle time, open receivables aging, revenue by customer/product, and delivery performance metrics.
- **EDI (Electronic Data Interchange) Integration:** Implement EDI 850 (Purchase Order) and EDI 810 (Invoice) to enable automated order exchange with large enterprise customers, reducing paper-based processes entirely.
- **SAP Fiori UX:** Replace traditional SAP GUI transactions with SAP Fiori apps (e.g., Manage Sales Orders app, Billing app) to provide a modern, mobile-friendly user interface for sales representatives on field.
- **Machine Learning for Price Optimization:** Integrate SAP with ML models to suggest optimal pricing based on customer history, market trends, and competitor pricing, feeding recommendations back into the SAP pricing conditions.

8. Conclusion

This capstone project successfully demonstrates the end-to-end Order-to-Cash (O2C) business process in SAP SD for TechNova Pvt. Ltd. Starting from enterprise structure configuration and master data setup, through the complete transactional cycle of Inquiry → Quotation → Sales Order → Delivery → Billing → Payment, the project covers all critical touchpoints of the O2C process.

The integration between SAP SD and SAP FI was a key highlight, demonstrating how a single billing transaction automatically creates accounting entries, eliminates dual data entry, and ensures financial accuracy. The project also explored advanced topics including credit management, pricing procedure configuration, and ATP availability checking.

Through this project, a deep understanding of how SAP drives operational efficiency, financial accuracy, and business process automation in a real enterprise environment has been developed. The O2C cycle remains the backbone of any product-selling company,

and SAP SD provides the most robust, configurable platform to execute and monitor it.

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3. SAP IMG (SPRO) – Implementation Guide for SD Enterprise Structure Configuration
4. SAP Transaction Codes Reference: VA01, VL01N, VF01, F-28, XD01, MM01
5. SAP S/4HANA Order-to-Cash Business Process Guide – SAP Press
6. Capstone Project Submission Guidelines – KIIT SAP Program (April 2026)